USE HAND, POWER AND MEASURING TOOLS

PRACTICAL MODULE

Module #	671101000-PM-01
NQF Level	level 3,
Notional hours	280
Credit(s)	Cr 28
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

INTRODUCTION

Congratulations on completing the program. As part of your training you are required to keep a logbook of all practical on the job training and exposure you receive during the learning process.

You will now be assigned to a mentor who will oversee your off-site training, usually referred to as on-the-job training.

The mentor will assist and advise you on the practical aspects of the job, how to fit into the company, what is expected of you as an employee and as a future supervisor.

Responsibilities of the learner include:

- One hundred percent commitment to the learning process. Learners are encouraged to study any additional source of information relevant to this learning process.
- Doing all assignments contained in this workbook as well any tasks and assignments received from your mentor or supervisor to whom you have been assigned.
- Although the mentor is responsible to sign off all sections completed, it is the learner's responsibility to ensure that all paperwork is completed and handed in for filing on his/her record of learning. It should be clearly stated to learners that a 100% complete record of learning, as prescribed by this logbook, is **their sole responsibility.** Any document missing from the record may result in your not being declared competent.
- Discuss any problems that you may have with your mentor.

Instructions to the Assessor

Congratulations on your appointment as a assessor to the learner. This is a very responsible assignment because you have been tasked with the responsibility of rounding off the learner's practical exposure.

You must ensure that you are familiar with all aspects of the work covered in this logbook because you must keep a daily account of the learner's performance.

You are also required to report to the learnership coordinator skills development facilitator, or as agreed between yourself, the coordinator and the learner regarding the learner's progress. Your responsibilities as mentor are as follows:

- Study the logbook and acquaint yourself with its content and format
- Remember this logbook is the learner's full record of learning and workplace exposure. All activities which the learner participates in must be recorded, and all documents produced in relation to this learnership must form part of the record of learning
- Get all the learners together and explain its purpose to them and also what is required of them
- Remember the mentor is the creator of learning and exposure opportunities. You should therefore not confine the learner's exposure to this logbook alone
- Continuously guide them in doing the assignments and arranging the planned exposure with the relevant departments
- File all duplicate records of learning on a file for each learner
- Send all original records to the training provider at the end of end of each month. The Seta also requires that copies be held at the companies
- This logbook need not be followed chronologically, but please note that the learner is required to work through the entire logbook by the end of the learning period

Provider Accreditation Requirements for the Module

Physical Requirements:

Demonstrate access to:

Electrical environment

categories of tools, equipment and processes in order for learners to be able to plan worksite set up for installing, wiring and connecting electrical equipment and control systems

Material include mild steel of 3mm for butt and lap welds , Butt-and lap welds on copper (maximum thickness of 8mm), Brazing is limited to mild steel only (maximum thickness of 3mm), Gas cutting is limited to mild steel only (maximum thickness of 6mm).

Human Resource Requirements:

Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting, commissioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within en electrical-related industry with a recognized Trade Certificate.

Trainer/learner ratio 1 to 12

Workshop coach/learner ratio 1 to 4

Legal Requirements:

Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Partner (AQP)

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ)

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA)

Critical Practical Activities to be Assessed Externally for the Module

None

Exemptions

PRACTICAL SKILL MODULE SPECIFICATIONS

List of Practical Skill Module Specifications

671101000-PM-01 Use hand, power and measuring tools,L3,Cr28

671101000-PS-02 Plan worksite set up for installing, wiring and connecting electrical equipment and control systems,L3, Cr5

671101000-PS-03 Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems,L3,Cr3

671101000-PS-04 Install wire-ways,L4, Cr5

671101000-PS-05 Install, wire and connect electrical equipment and control systems, L4,Cr38

671101000-PS-06 Conduct pre-commission inspection (power off) and test new and existing installations, L4,Cr5

671101000-PS-07 Carry out commissioning tests,L4,Cr13

671101000-PS-08 Fault find and repair electrical control systems and electrical installations, L4, Cr22

SECTION A-Guidelines for Practical Skills
PM-01-PS01: Select, use and care of engineering hand tools

Scope of Practical Skill

Given work instructions, checklists, work area, drawings, documents, templates, forms, safety and quality principles, hand saws, hammers, screw drivers, sockets, spanners, chassis punches, side cutters, pliers, wire strippers, drill bits, measuring and marking off tools, fastening tools, standard operating procedures and, statutory requirements, the learner should be able to:

PA0101 Select engineering hand tools for serviceability. PA0102 Use engineering hand tools.

PA0103 Care for engineering hand tools.

Applied Knowledge

AK0101 Workshop procedures including housekeeping practices according to statutory requirements.

AK0102 Techniques of maintaining hand tools

AK0103 Techniques to correctly store hand tools

OBSERVATION EVALUATION

NAME: ..... SURNAME: ..... COMPANY..... ID.....	**EVALUATION CHECKLIST** DATE: TIME:

EVALUATION CRITERIONS	**MET REQUIREMENTS**	**DID NOT MEET REQUIREMENTS**	**COMMENTS OR ACTION REQUIRED**
ABILITY OF THE LEARNER TO:			
1. Identify Unsafe and faulty tools and mark for repair/maintenance or replacement according to set procedures			
2. Use the correct application of tools to measure and mark out steel, copper, aluminium, wood, plastics and synthetic materials.			
3. Use the correct application of tools for cutting steel, copper,			

8. Lubricate the adjusting/moving mechanisms of shifting spanners, gas pump pliers, vice grip, dividers, crimping tools, stilson wrench to ensure free movement and serviceability.			
9. Clean Engineering files cutting edge using a wire brush to ensure serviceability. Scribers, dividers and centre punches were sharpened to ensure serviceability.			

GENERAL COMMENTS OF OBSERVER:

Date.....	Time started.....	Time completed.....
ASSESSOR NAME	**ASSESSOR SIGNATURE**	**LEARNER (SIGNATURE)** .....

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Select Power tools according to the task requirements.
2. Cary out Pre-operational check in terms of safety requirements and function.
3. Recognise Hazards associated with the use of power tools and take necessary precautions according to workshop procedures.
4. Identify and report Faulty and unsafe tools in accordance with standard operating procedures
5. Select and use Personal safety equipment according to tool requirement.
6. Select and set up Fixed and portable power tools according to job requirements.
7. Utilise all guards and securing mechanisms in terms of job requirements.
8. Use Fixed and portable power tools safely and in accordance with manufacturer's specifications.
9. Inspect, clean and lubricate Power tools after use according to workshop practices and/or manufactures specifications.
10. Secure and repair loose items and minor defects in accordance to workshop procedures.
11. Store Portable power tools and accessories in accordance to workshop procedures.

standard operating procedures			
5. Select and use Personal safety equipment according to tool requirement.			
6. Select and set up Fixed and portable power tools according to job requirements.			
7. Utilise all guards and securing mechanisms in terms of job requirements.			
8. Use Fixed and portable power tools safely and in accordance with manufacturer's specifications.			
9. Inspect, clean and lubricate Power tools after use according to workshop practices and/or manufactures specifications.			
10. Secure and repair loose items and minor defects in accordance to workshop procedures.			
11. Store Portable power tools and accessories in accordance to workshop procedures.			

SECTION C-Guidelines for Practical Skills

PM-01-PS03: Select, use and care for electrical measuring instruments

Scope of Practical Skill

Given work instructions, work area, documents, multimeter (ohms, volts, amp`s), insulation tester, clip on ammeter, earth leakage polarity tester, oscilloscope standard operating procedures and, statutory requirements the learner should be able to:

PA0301 Identify and select portable electrical measuring instruments.

PA0302 Use and interpret portable electrical measuring instrument readings.

PA0303 Care for portable electrical measuring instruments.

Applied Knowledge

AK0301 Handling procedures related to portable measuring instrument.

AK0302 Basic operating principles of portable electrical measuring instruments.

AK0303 Types, applications and functions of portable electrical measuring instruments.

OBSERVATION EVALUATION

NAME: ..... SURNAME: ..... COMPANY..... ID.....	**EVALUATION CHECKLIST** DATE: TIME:

EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Interpret Job instructions correctly and sequence of operation is determined.			
2. Select Portable electrical measuring instruments correctly according to workshop procedures and to meet safety and job requirements			
3. Identify and mark Unsafe and faulty measuring instruments for repair or replacement.			
4. Read Portable measuring instruments correctly and			

Date.....	Time started.....	Time completed.....
ASSESSOR NAME	**ASSESSOR SIGNATURE**	**LEARNER (SIGNATURE)** .....

INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Inspect Work area for safe working conditions and corrective action is taken where required.
2. Select Applicable soldering equipment as required by task.
3. Correct select Soldering material as required by the task.
4. Clean all connections from any dirt or oxidation.
5. Tinning of wire and connections was done according to manufacturer's specifications.
6. Solder Connections according to set specifications and techniques.
7. Lay out Components on the circuit board according to the circuit diagram.
8. Solder Components according to soldering techniques.

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GENERAL COMMENTS OF OBSERVER:		
Date.....	Time started.....	Time completed.....
ASSESSOR NAME	**ASSESSOR SIGNATURE**	**LEARNER (SIGNATURE)** .....

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

1. Obtain and inspect the applicable safety equipment as per safe work procedures and statutory requirements.
2. Set up the electric arc welding equipment and check cable connections for tightness to ensure effective operation as per safe work procedures.
3. Correct apply the cleaning materials for preparing the metal surfaces to be welded as per task requirements.
4. Wear the appropriate personal protective equipment as per safe work procedures.
5. Bevel the edges of the metals welded correctly where needed as per specific task requirements.
6. Clean the contact surfaces of the earth clamp, metals earthed and the electrode holder to ensure proper current flow during the welding process.
7. Correctly select the required current setting on the welding machine for electric arc welding according to application requirements.
8. Carry out the joining of metals by means of electric arc welding was and position correct electrode positioning and movement according to application requirements.
9. Cool down Hot metal surfaces and remove all slag from the welding joints as per safe work procedures.
10. Correctly store Electric arc welding machine, equipment and materials as per applicable safety standards.
11. Clean the work area and dispose waste materials according to local housekeeping standards.
12. Clean and store Safety equipment according to statutory requirements.

OBSERVATION EVALUATION

NAME:	**EVALUATION CHECKLIST**

5. Bevel the edges of the metals welded correctly where needed as per specific task requirements.			
6. Clean the contact surfaces of the earth clamp, metals earthed and the electrode holder to ensure proper current flow during the welding process.			
7. Correctly select the required current setting on the welding machine for electric arc welding according to application requirements.			
8. Carry out the joining of metals by means of electric arc welding was and position correct electrode positioning and movement according to application requirements.			
9. Cool down Hot metal surfaces and remove all slag from the welding joints as per safe work procedures.			
10. Correctly store Electric arc welding machine,			

PM-01-PS06: Carry out basic gas cutting in an electrical environment

Scope of Practical Skill

Given work instructions, work area, documents, Oxygen-and acetylene gas cylinders and keys, gas regulators and flashback arrestors, rubber gas hoses and clamps, cutting torches and nozzles, nozzle cleaning equipment, spark lighter, , metal surface cleaning materials (wire brush, emery cloth, chemicals), gas welding goggles, leather aprons and boot protectors, safety gloves and overalls the learner should be able to:

PA0601 Prepare for gas cutting.

PA0602 Apply basic gas cutting process.

Applied Knowledge

AK0601 Names, applications and functions of gas cutting equipment.

AK0602 Handling procedures related to gas cutting equipment.

AK0604 Safe work procedures and workshop practices.

AK0605 Workshop procedures including housekeeping practices according to statutory requirements.

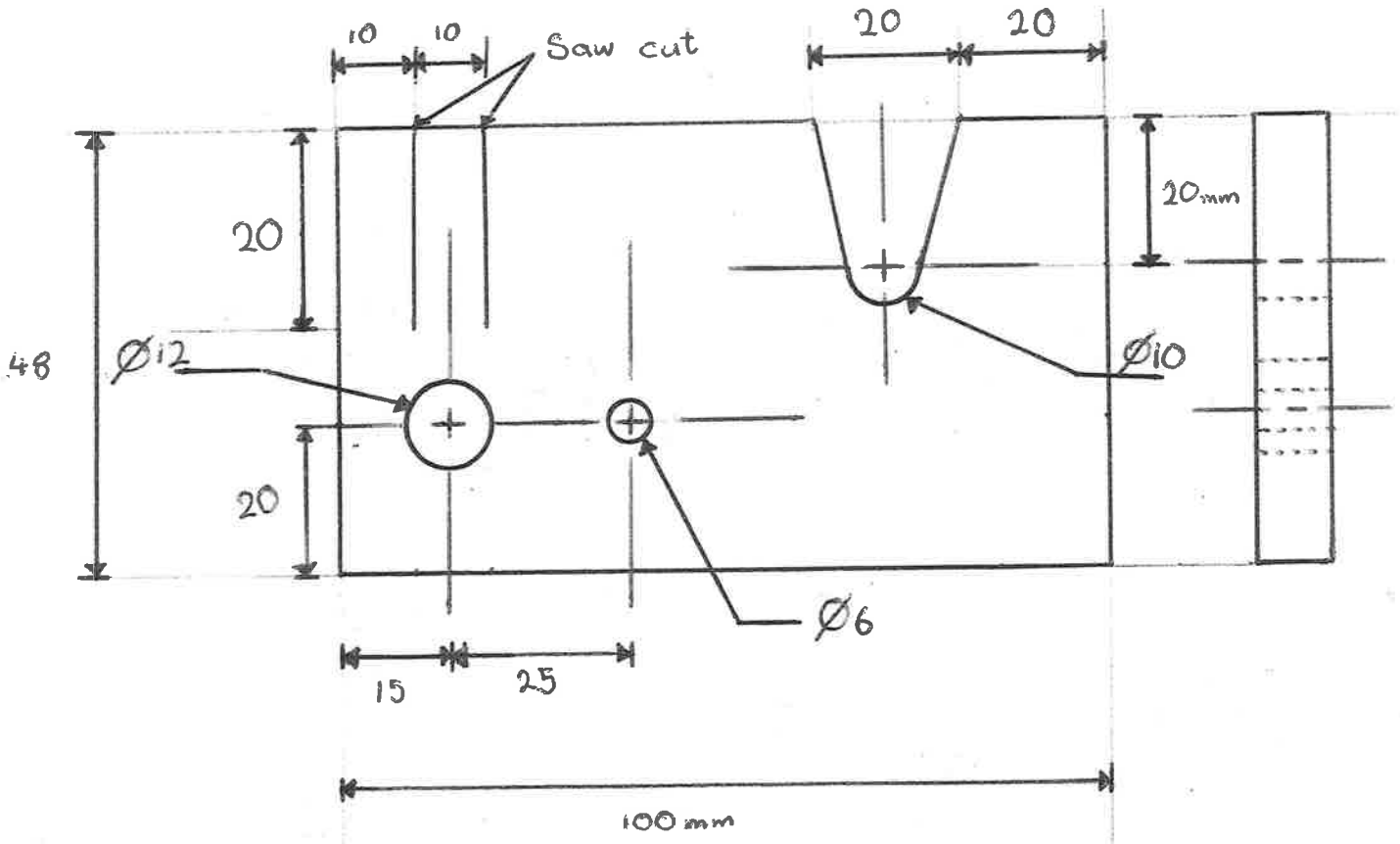
INSTRUCTIONS

You are required to perform the task below and Your Supervisor/facilitator/mentor will assess you based on the below observation checklist attached:

ID.....	TIME:

EVALUATION CRITERIONS	MET REQUIREMENTS	DID NOT MEET REQUIREMENTS	COMMENTS OR ACTION REQUIRED
ABILITY OF THE LEARNER TO:			
1. Obtain and inspect the applicable safety equipment as per safe work procedures and statutory requirements.			
2. Set up the gas cutting equipment and check for gas leaks and correct cylinder pressures according to safe work procedures and industry requirements.			
3. Prepare the relevant cutting material limited to mild steel only (maximum thickness of 6mm), required for the task to be ready and available, and select nozzles correctly to match the application.			
4. Handle the gas cylinders were handled safely as per safe work procedures			

equipment and materials as per applicable safety standards.			
12. Clean the work area and dispose waste materials according to local housekeeping standards.			
GENERAL COMMENTS OF OBSERVER:			
Date……..	Time started……..	Time completed……..	
ASSESSOR NAME ……	**ASSESSOR SIGNATURE** ……	**LEARNER (SIGNATURE)** ……	


10
10
Saw cut
20
20
20mm
48
20
Ø12
Ø10
20
Ø6
15
25
100 mm